

01 | Agentic Workflow

02 | Project

Topic: Waste in the city

The goal is to study how different types of agents and infrastructure influence the spatial and temporal distribution of waste in the city.

- The city is modeled as a grid world consisting of:
 - ▶ streets / paths where agents can move
 - ▶ walls / buildings that block movement
 - ▶ public areas where waste may accumulate
 - ▶ fixed waste infrastructure, such as bins and containers

Your model should include at least the following entities:

1. Local humans

- ▶ move according to regular daily patterns
- ▶ generate small amounts of waste with a given probability
- ▶ may use nearby bins if available

2. Tourists

- ▶ move less predictably
- ▶ prefer central or attractive areas
- ▶ may generate more waste in crowded areas

3. Cleaning service

- ▶ move through streets
- ▶ detect and collect waste from public spaces and bring to the disposal areas, where waste is processed/terminated
- ▶ may follow simple rule-based strategies, e.g. nearest waste, random patrol, or fixed route

4. Dust bins and dust containers

- ▶ fixed infrastructure
- ▶ have limited capacity
- ▶ receive waste from humans and robots
- ▶ may overflow if not emptied

5. Dust transporters

- ▶ collect waste from full bins or containers
- ▶ transport it to a disposal point outside or at the edge of the city
- ▶ follow street paths and cannot pass through buildings

How do city structure, movement patterns, bin placement, cleaning strategies, and transporter frequency affect waste accumulation in a city?

ABM task

The main interest is not one agent's decision process, but the **emergent city-level waste distribution caused by many interacting agents over time.**

1. Generate a city with a given structure
2. Run multiple scenarios:
 - ▶ Many streets/Many buildings
 - ▶ Few bins vs. many bins
 - ▶ Random robot patrol vs. nearest-waste strategy
 - ▶ Low tourist density vs. high tourist density
 - ▶ Rare transporter visits vs. frequent transporter visits
3. Measure and visualize statistics over simulation steps
 - ▶ total waste on streets
 - ▶ number of overflowing bins
 - ▶ average waste per district / city region
 - ▶ robot cleaning efficiency
 - ▶ transporter workload
4. Implement your simulation in Mesa or AgentPy
5. Use graph search algorithms where it makes sense

Creative Extension Requirement (Mandatory)

In addition to the core implementation, each group must

- design and
- implement **an original extension that meaningfully enhances the model**

Repetition

If two or groups will have **identical** 'creative extensions', the evaluation will be reduced for these groups (as no creative solution is implemented!)